Programming in Python

Digital Business and Data Science
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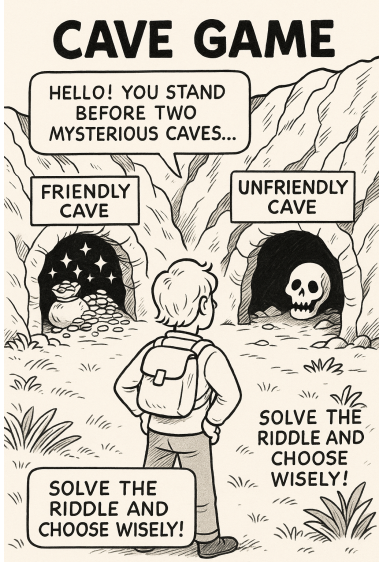
THE CAVE GAME – BUILD YOUR OWN ADVENTURE USING FUNCTIONS

Learning Goal
In this exercise, you will develop your own mini text game using Python functions.
You will practice how to:

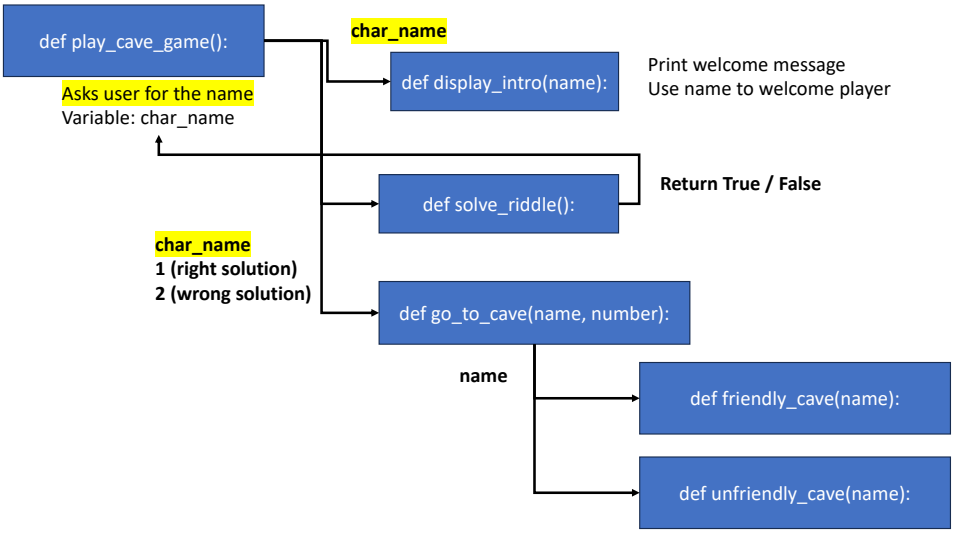
- Define and call functions
- Pass information between functions
- Structure your code clearly and logically



The Scenario
You are standing in front of two mysterious caves.
One cave is friendly and full of treasure,
the other is dark and dangerous.
Your mission: solve a riddle and choose wisely!

Your Task
You will build the game step by step using functions.
The visualization of the program structure provided in class shows how the different functions are connected.

Function Overview



Function Name: play_cave_game()
Purpose: The main function that controls the game flow
Hint: Starts everything

Function Name: display_intro(name)
Purpose: Prints a welcome message using the player's name
Hint: Use print() and string concatenation

Function Name: solve_riddle()
Purpose: Asks the player a riddle and returns whether the answer is right
Hint: Use input() and if / else

Function Name: go_to_cave(name, number)
Purpose: Sends the player to the right or wrong cave
Hint: Use an if condition to decide

Function Name: friendly_cave(name)
Purpose: Prints a message for the good cave
Hint: Keep it fun and creative

Function Name: unfriendly_cave(name)
Purpose: Prints a message for the bad cave
Hint: Add a bit of drama

Steps to Develop

1. **Start the main function**
   - Ask the user for their character name.
   - Store it in a variable, for example char_name.
2. **Display the introduction**
   - Create a separate function that greets the player and sets the scene.
   - It should use the name you just asked for.
3. **Ask the riddle**
   - The riddle should have a clear, numeric answer to make checking easy.
   - Think about how to make the program know whether the answer is correct.
   - The function should return True if correct or False if wrong.
4. **Choose the cave**
   - The main function should check the result of the riddle.
   - If it is True, send the player to a friendly cave.
   - If it is False, send them to an unfriendly cave.
5. **Create the two cave functions**
   - Each should print a short message.
   - Use the player's name inside the message.

Challenge Ideas (optional)

- Add more caves or riddles
- Randomize which cave is friendly
- Give the player multiple chances
- Add a restart or "play again" option

Reflection
After finishing, discuss or write briefly:

1. What did you find easiest or hardest about using functions?
2. What happens when you forget to pass a variable (like name) to another function?
3. How do return values help control the flow of your program?